

Application of TSVMs in bankruptcy prediction for credit risk

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Recent years have seen an explosive growth in computing power and data storage within business organizations. From a business perspective, this means that most companies now have massive archives of customer and product data and more often than not these archives are far too large for human analysis. An obvious question has therefore arisen, “How can one turn these immense corporate data archives to commercial advantage?” To this end, a number of common applications have arisen, from predicting which products a customer is most likely to purchase, to designing the perfect product based on responses to questionnaires. The theory and development of these processes has grown into a discipline of its own, known as Data Mining, which draws heavily on the related fields of Machine Learning, Pattern Recognition, and Mathematical Statistics.

The Data Mining discipline is still developing however, and a great deal of suboptimal and ad hoc analysis is being done. This is partly due to the complexity of the problems, but is also due to the vast number of available techniques. Even the most fundamental task in Data Mining – that of inductive inference, or making predictions based on examples, can be tackled by a great many different techniques. Some of these techniques are very difficult to tailor to a specific problem and require highly skilled human design; others are more generic in application and can be treated more like the proverbial “black box”. One particularly generic and powerful method, known as the Support Vector Machine (SVM) has proven to be both easy to apply and capable of producing results that range from good to excellent in comparison to other methods. While application of the method is relatively straightforward, the practitioner can still benefit greatly from a basic understanding of the underlying machinery.

SVMs have developed based on principal of minimizing structural errors from statistical learning theory. These classifiers have better self-efficiency than other methods. One of the most important factors for a good classifier is the ability to work with least training data. In real world, there are not enough training data for learner or preparing it is time-consuming and difficult. TSVM is a special SVM that is efficient and its goal is classifying by minimum number of training data. So in this paper we will bring it into play.

In addition to their solid mathematical foundation in statistical learning theory, SVMs have demonstrated highly competitive performance in numerous real-world applications, such as medical diagnosis, bioinformatics, face recognition, image

processing and text mining, which has established SVMs as one of the most popular, state-of-the-art tools for knowledge discovery and data mining. SVMs possess the well-known ability of being universal approximators of any multivariate function to any desired degree of accuracy. Therefore, they are of particular interest to modeling highly nonlinear, complex systems and processes.

Bankruptcy is the condition in which a business cannot meet its debt obligations and petitions for either reorganization of its debts or liquidation of its assets. In the action, the property of a debtor is taken over by a receiver or trustee in bankruptcy for the benefit of the creditors. This action may be voluntary or involuntary. Signs of potential financial distress are evident long before bankruptcy occurs. Financial distress begins when a corporation is unable to meet its scheduled payments or when the projection of future cash flows points to an inability to do so in the near future.

The prediction of corporate bankruptcies is an important and widely studied topic since it can have significant impact on bank lending decisions and profitability. This work presents two contributions. First we review the topic of bankruptcy prediction, with emphasis on svm models. Second, we develop a TSVM bankruptcy prediction model.

Keywords

Transductive support vector machines, bankruptcy prediction, credit risk,

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